* The documentation and process conversion *
* measures necessary to comply with this revision *
* revision shall be completed by 13 December 1996. *

* INCH POUND *

MIL-PRF-19500/596D
13 September 1996
SUPERSEDING
MIL-S-19500/596C
7 April 1995

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, REPETITIVE AVALANCHE, FIELD EFFECT,
TRANSISTOR, N-CHANNEL, SILICON,
TYPES 2N7218, 2N7219, 2N7221, 2N7222, 2N7218U, 2N7219U,
2N7221U, 2N7222U, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a N-channel, enhancement mode, MOSFET, power transistor intended for use in high density power switching applications. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500, with avalanche energy maximum ratings (E_{AR} and E_{AS}) and maximum avalanche current I_{AR}. Two levels of product assurance for die (element evaluation).

1.2 Physical dimensions. See figure 1 (T0-254AA), figure 2 for JANHC and JANKC (die) dimensions, and figure 3 for surface mount.

1.3 Maximum ratings.

Type	$V_{(BR)DSS}$ min	P_T 1/	P_T	V_{GS}	I_{D1} 2/	I_{D2} 2/	I_S	I_{DM}	T_{op} and T_{STG}
5/	V_{GS} = 0 V I_D = 1.0 mA dc	T_C = 25°C	T_A = 25°C		T_C = 25°C	T_C = 100°C		3/	
	V dc	W	W	V dc	A dc	A dc	A dc	A (pk)	°C
2N7218	100	125	4	±20	28	20	28	112	-55
2N7219	200	"	"	"	18	11	18	72	to
2N7221	300	"	"	"	10	6	10	40	+150
2N7222	400	"	"	"	8	5	8	32	

See footnotes next page.

* Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this *
* document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad St., *
* Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) *
* appearing at the end of this document or by letter. *

AMSC N/A FSC 5961
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.

Type 5/	I_{AR} 2/	E_{AS}	E_{AR}	$V_{(ISO)}$ at 70,000 feet	$R_{\Theta JC}$ max	$r_{DS(on)}$ max 1/ 4/ V_{GS} = 10 V $I_D = I_{D2}$	
						T_J = 25°C	T_J = 150°C
	A	mj	mj		°C/W	Ω	Ω
2N7218	28	250	12.5		1.0	0.077	0.154
2N7219	18	450	"		"	0.18	0.387
2N7221	10	650	"	400	"	0.55	1.32
2N7222	8	700	"	500	"	0.85	2.04

1/ Derate linearly 1.0 W/°C for $T_C > 25°C$; $P_T = \frac{T_{J(max)} - T_C}{R_{\Theta JX}}$

2/ $I_D = \sqrt{\frac{T_{J(max)} - T_C}{(R_{\Theta JX}) \times (R_{DS(on)} \text{ at } T_{J(max)})}}$

3/ I_{DM} = 4 x I_D as calculated by footnote 2/.

4/ Pulsed (see 4.5.1).

5/ Electrical characteristics for "U" suffix devices are identical to the corresponding non "U" devices unless otherwise specified.

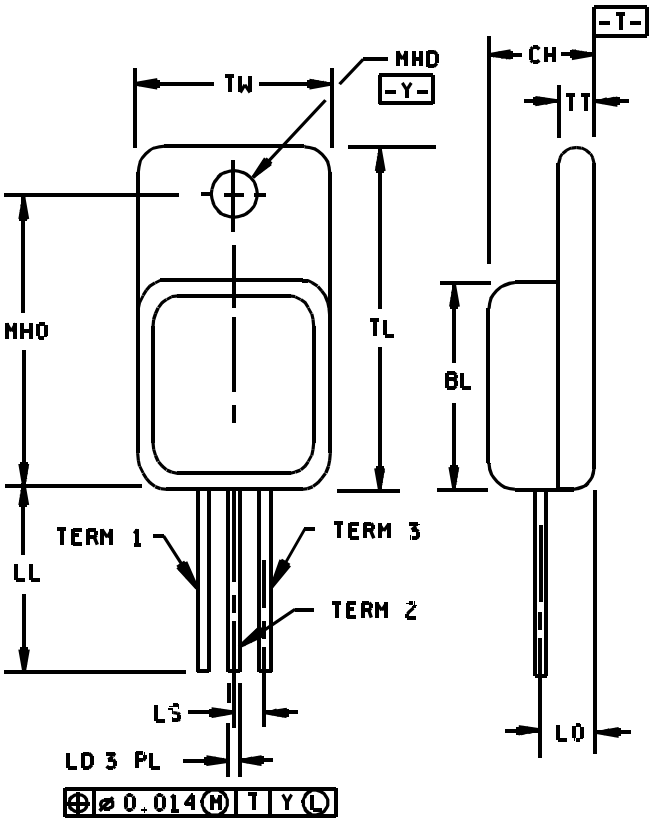
1.4 Primary electrical characteristics. Unless otherwise specified, T_C = +25°C.

Type 1/	$V_{(BR)DSS}$ min V_{GS} = 0 V I_D = 1.0 mA dc	$V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ I_D = 0.25 mA		I_{DSS1} max V_{GS} = 0 V V_{DS} = 80 percent of rated V_{DS}	$r_{DS(on)1}$ V_{GS} = 10 V $I_D = I_{D2}$
	V dc	V dc		μA dc	ohms
		Min	Max		
2N7218	100	2.0	4.0	25	0.077
2N7219	200	2.0	4.0	25	0.18
2N7221	400	2.0	4.0	25	0.55
2N7222	500	2.0	4.0	25	0.85

1/ Electrical characteristics for "U" suffix devices are identical to the corresponding non-"U" suffix devices.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.

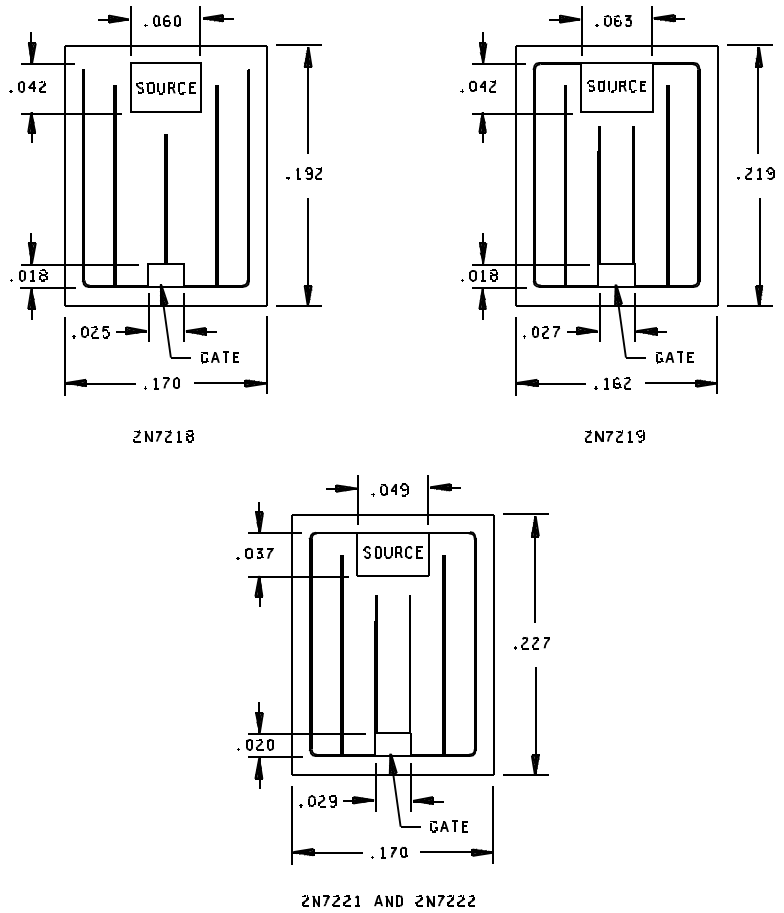
Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CH	.249	.260	6.32	6.60	
TT	.040	.050	1.02	1.27	
LO	.150 BSC		3.81 BSC		
TL	.790	.800	20.07	20.32	3, 4
BL	.535	.545	13.59	13.84	
LS	.150 BSC		3.81 BSC		
TW	.535	.545	13.59	13.84	3, 4
LL	1.195	1.235	30.35	31.37	
MHO	.665	.685	16.89	17.40	
MHD	.139	.149	3.53	3.78	
LD	.035	.045	0.89	1.14	
Term 1	Drain				
Term 2	Source				
Term 3	Gate				



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Glass meniscus included in dimension D and E.
4. All terminals are isolated from the case.
5. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions for TO-254AA.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.



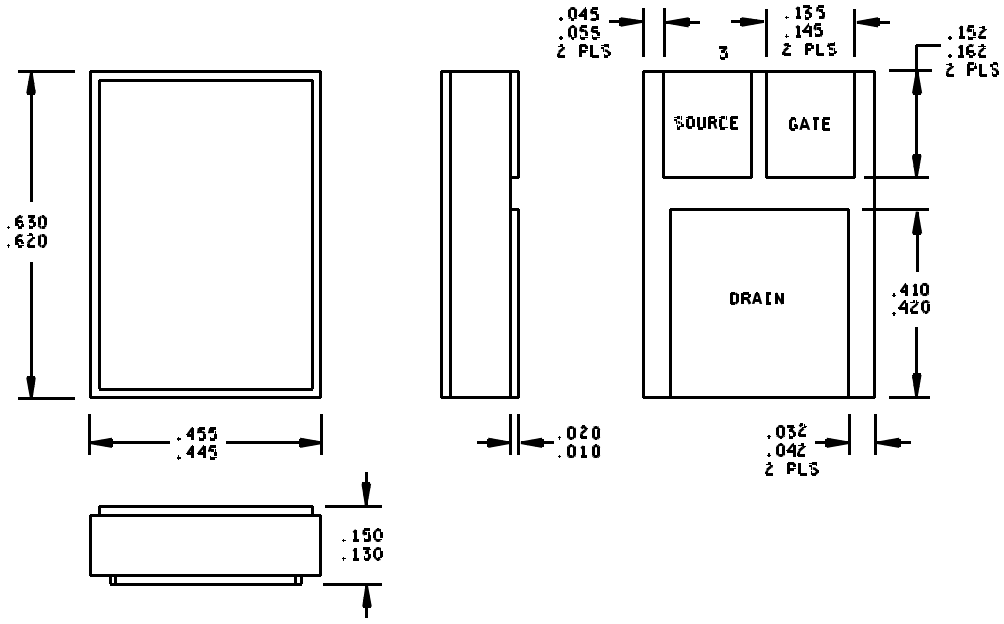
Inches	mm	*	Inches	mm	*	Inches	mm
.018	0.46	*	.037	0.94	*	.162	4.11
.020	0.51	*	.042	1.07	*	.170	4.32
.025	0.64	*	.049	1.24	*	.192	4.88
.027	0.69	*	.060	1.52	*	.219	5.56
.029	0.74	*	.063	1.60	*	.227	5.77

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).
4. The physical characteristics of the die thickness are .0187 inch (0.475 mm). The back metals are chromium, nickel, and silver. The top metal is aluminum and the back contact is the drain.

FIGURE 2. JANHC and JANKC (A-version) die dimensions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.



Inches	mm	*	Inches	mm
.010	0.25	*	.150	3.81
.020	0.51	*	.152	3.86
.032	0.81	*	.162	4.11
.042	1.07	*	.410	10.41
.045	1.14	*	.420	10.67
.055	1.40	*	.445	11.30
.130	3.30	*	.455	11.56
.135	3.43	*	.620	15.75
.145	3.68	*	.630	16.00

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.

FIGURE 3. Dimensions and configuration of surface mount package outline.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated performance specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-19500.

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 herein. Methods used for electrical isolation of the terminal feedthroughs shall employ materials that contain a minimum of 90 percent AL_2O_3 (ceramic). Examples of such construction techniques are metallized ceramic eyelets or ceramic walled packages.

3.3.1 Lead formation, material, and finish. Lead material shall be Kovar or Alloy 52; a copper core or plated core is permitted. Lead finish shall be solderable in accordance with MIL-STD-750 and MIL-PRF-19500. Where a choice of finish is desired, it shall be specified in the acquisition document (see 6.2). When lead formation is performed, as a minimum, the vendor shall perform 100 percent hermetic seal in accordance with screen 14 and 100 percent dc testing in accordance with group A, subgroup 2 herein.

3.3.2 Internal construction. Multiple chip construction shall not be permitted.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500, except at the option of the manufacturer, the country of origin and/or the manufacturers identification may be omitted from the body of the transistor.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.

3.5.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. The following handling procedures shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source. R $\leq$ 100 k, whenever bias voltage is to be applied drain to source.

3.6 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I.

3.8 Electrical test requirements. The electrical test requirements shall be the subgroups specified in paragraphs 4.4.2 and 4.4.3.

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, and as specified herein. Alternate flow is allowed for qualification inspection in accordance with figure 2 of MIL-PRF-19500.

4.2.1 Group E inspection. Group E inspection shall be in accordance with table II herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.

4.3 Screening (JANS, JANTX, and JANTXV levels). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
1/	Gate stress test (see 4.5.5)	Gate stress test (see 4.5.5)
1/ 2/	Method 3470 (see 4.5.4)	Method 3470 (see 4.5.4)
1/	Method 3161 (see 4.5.3)	Method 3161 (see 4.5.3)
9 1/	I_{GSS1}, I_{DSS1}, subgroup 2 of table I herein	Subgroup 2 of table I herein
10	Method 1042, test condition B	Method 1042, test condition B
11	I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$ subgroup 2 of table 1 herein; ΔI_{GSS1} = ±20 nA dc or 100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 μA dc or ±100 percent of initial value, whichever is greater.	I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$, subgroup 2 of table I herein.
12	Method 1042, test condition A	Method 1042, test condition A or T_A =175°C and t = 48 hours.
13	Subgroups 2 and 3 of table I herein; ΔI_{GSS1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 μA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)(1)}$ = ±20 percent of initial value. $\Delta V_{GS(th)1}$ = ±20 percent of initial value.	Subgroup 2 of table I herein; ΔI_{GSS1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 μA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)(1)}$ = ±20 percent of initial value. $\Delta V_{GS(th)1}$ = ±20 percent of initial value.

1/ Shall be performed anytime before screen 10.
2/ This is a stress test designed to ensure a rugged product.

4.3.1 Screening (JANHC and JANKC). Screening of die shall be in accordance with MIL-PRF-19500 appendix H. As a minimum Die, shall be 100 percent probed per group A, subgroup 2 except test current shall not exceed 20 amperes.

4.3.2 JANHC and JANKC die. Qualification shall be in accordance with MIL-PRF-19500.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein. Electrical measurements (end-points) shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JANTX and JANTXV) of MIL-PRF-19500 and herein. Electrical measurements (end-points) shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Conditions
3	1051	Test condition G.
3	2037	Test condition A. All internal bond wires for each device shall be pulled separately.
4	1042	Test condition D; the heating cycle shall be 1 minute minimum, 2,000 cycles. No heat sink nor forced air cooling on the device shall be permitted during the cycle.
5	1042	Accelerated steady-state operation life; test condition A, V_{DS} = rated T_A = +175°C, t = 120 hours minimum. Read and record $V_{(BR)DSS}$ (pre and post at 1 mA = I_D. Read and record I_{DSS} (pre and post). Deltas for $V_{(BR)DSS}$ shall not exceed 10% and I_{DSS} shall not exceed 25 µA. Accelerated steady-state gate stress; condition B, V_{GS} = rated, T_A = +175°C, t = 24 hours.
5	2037	Bond strength (Al-Au die interconnects only); test condition A.
6	3161	See 4.5.2.

4.4.2.2 Group B inspection, table VIb (JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
2	1051	Test condition G.
3	1042	Test condition D, 2,000 cycles minimum. The heating cycle shall be 1 minute minimum.
4	2075	See 3.3.2.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
2	1056	Test condition B.
2	2036	Tension: Test condition A; weight = 10 lbs, t = 10 s (not applicable to "U" suffix version).
6	1042	Test condition D, 6,000 cycles minimum. The heating cycle shall be 1 minute minimum.

4.5 Methods of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in MIL-STD-750.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.

4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit of $R_{\Theta JC(max)}$ shall be 1.0°C/W . The following parameter measurements shall apply:

a. Measuring current (I_M): 10 mA.
b. Drain heating current (I_H): 3.3 A
c. Heating time (t_H): Steady state (see MIL-STD-750, method 3161 for definition).
d. Drain-source heating voltage (V_H): 25 V minimum.
e. Measurement time delay (t_{MD}): 30 to 60 μs maximum.
f. Sample window time (t_{SW}): 10 μs maximum.

4.5.3 Thermal impedance ($Z_{\Theta JC}$ measurements). The $Z_{\Theta JC}$ measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit (not to exceed the group A, subgroup 2 limit or figure 4 thermal impedance curve) for $Z_{\Theta JC}$ in screening (table IV of MIL-PRF-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for Engineering evaluation and disposition. This procedure may be used in lieu of an in line monitor. The following parameter measurements shall apply:

a. Measuring current (I_M): 10 mA.

b. Drain heating current (I_H): 3.3 A.

c. Heating time (t_H): 100 ms.

d. Drain-source heating voltage (V_H): 25 V.

e. Measurement time delay (t_{MD}): 30 to 60 μs maximum.

f. Sample window time (t_{SW}): 10 μs maximum.

4.5.4 Single pulse avalanche energy (E_{AS}).

a. Peak current (I_{AS}): I_{D1}.

b. Peak gate voltage (V_{GS}): 10 V.

c. Gate to source resistor (R_{GS}): $25 \leq R_{GS} \leq 200\ \Omega$.

d. Initial case temperature: +25°C +10°C, -5°C.

e. Inductance: $(2\ E_{AS}/(I_{D1})^2)((V_{BR} - V_{DD})/V_{BR})$ mH minimum.

f. Number of pulses to be applied: 1 pulse minimum.

g. Supply voltage (V_{DD}): 50 V, 25 V for 100 V devices.

4.5.5 Gate stress test.

V_{GS} = 30 V minimum.

t = 250 μs minimum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Breakdown voltage, drain to source	3407	V_{GS} = 0 V dc, I_D = 1.0 mA, Bias condition C	$V_{(BR)DSS}$			
2N7218				100		V dc
2N7219				200		V dc
2N7221				400		V dc
2N7222				500		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$ I_D = 0.25 mA	$V_{GS(th)1}$	2.0	4.0	V dc
Gate current	3411	V_{GS} = +20 V dc V_{DS} = 0 V dc Bias condition C	I_{GSSF1}		100	nA dc
Gate current	3411	V_{GS} = -20 V dc V_{DS} = 0 V dc Bias condition C	I_{GSSR1}		-100	nA dc
Drain current	3413	V_{GS} = 0 V dc Bias condition C V_{DS} = 80% of rated V_{DS}	I_{DSS1}		25	µA dc
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc Condition A pulsed (see 4.5.1) I_D = rated I_{D2} (see 1.3)	$r_{DS(on)1}$			
2N7218					0.077	Ω
2N7219					0.18	Ω
2N7221					0.55	Ω
2N7222					0.85	Ω
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc Condition A pulsed (see 4.5.1) I_D = rated I_{D1} (see 1.3)	$r_{DS(on)2}$			
2N7218					0.125	Ω
2N7219					0.25	Ω
2N7221					0.70	Ω
2N7222					0.95	Ω

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward voltage (source drain diode)	4011	V_{GS} = 0 V dc I_D = rated I_{D1} (see 1.3)	V_{SD}			
2N7218					1.5	V
2N7219					1.5	V
2N7221					1.5	V
2N7222					1.5	V
Subgroup 3						
High temperature operation:		$T_C = T_J$ = +125°C				
Gate current	3411	Bias condition C V_{GS} = 20 V dc V_{DS} = 0 V dc	I_{GSSF2}		200	nA dc
Gate current	3411	Bias condition C V_{GS} = -20 V dc V_{DS} = 0 V dc	I_{GSSR2}		-200	nA dc
Drain current	3413	Bias condition C V_{GS} = 0 V dc V_{DS} = 100% of rated V_{DS}	I_{DSS2}		1.0	mA dc
		V_{DS} = 80% of rated V_{DS}	I_{DSS3}		0.25	mA dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$ I_D = 0.25 mA	$V_{GS(th)2}$	1.0		V dc
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc pulsed (see 4.5.1) $I_D = I_{D2}$ (see 1.3)	$r_{DS(on)3}$			
2N7218					0.24	Ω
2N7219					0.48	Ω
2N7221					1.44	Ω
2N7222					2.04	Ω
Low temperature operation:		$T_C = T_J$ = -55°C				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$ I_D = 0.25 mA	$V_{GS(th)3}$		5.0	V dc

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Switching time test	3472	I_D = rated I_{D2} V_{GS} = 10 V dc pulsed Gate drive impedance = 9.1 Ω V_{DD} = 50% of $V_{BR(DSS)}$				
Turn-on delay time			$t_{d(on)}$			
2N7218					21	ns
2N7219					20	ns
2N7221					25	ns
2N7222					21	ns
Rise time			t_r			
2N7218					105	ns
2N7219					105	ns
2N7221					92	ns
2N7222					73	ns
Turn-off delay time			$t_{d(off)}$			
2N7218					64	ns
2N7219					58	ns
2N7221					79	ns
2N7222					72	ns
Fall time			t_f			
2N7218					65	ns
2N7219					67	ns
2N7221					58	ns
2N7222					51	ns
Subgroup 5						
Safe operating area test (high voltage)	3474	See figure 5 t_p = 10 ms V_{DS} = 80 percent of rated $V_{BR(DSS)}$ V_{DS} = 200 V dc maximum				
Electrical measurements		See table III, steps 1, 2, 3, 4, 5, 6, and 7				
Subgroup 6						
Not applicable						

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 7						
Gate charge	3471	Condition B				
On-state gate charge			$Q_{g(on)}$			
2N7218					59	nC
2N7219					60	nC
2N7221					65	nC
2N7222					68.5	nC
Gate to source charge	3053		Q_{gs}			
2N7218					12	nC
2N7219					10.6	nC
2N7221					10	nC
2N7222					12.5	nC
Gate to drain charge			Q_{gd}			
2N7218					30.7	nC
2N7219					37.6	nC
2N7221					40.5	nC
2N7222					42.4	nC
Reverse recovery time	3473	di/dt = 100 A/μs	t_{rr}			
		$V_{DD} \leq 30$ V				
2N7218		I_D = rated I_{D1} (see 1.3)			400	ns
2N7219					500	ns
2N7221					600	ns
2N7222					700	ns

1/ For sampling plan, see MIL-PRF-19500.
2/ Unless otherwise specified, inspections, conditions, and limits for "U" suffix devices are identical to non-"U" suffix devices.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.

TABLE II. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Qualification inspection
	Method	Conditions	
Subgroup 1			45 devices, c = 1
Temperature cycling	1051	-55°C to 150°C, 500 cycles	
Hermetic seal	1071		
Fine leak			
Gross leak			
Electrical measurements		See table I, group A, subgroup 2	
Subgroup 2 1/			45 devices, c = 1
Steady-state reverse bias	1042	Condition A; 1000 hours	
Electrical measurements		See table I, group A, subgroup 2	
Steady-state gate bias	1042	Condition B; 1,000 hours	
Electrical measurements		See table I, group A, subgroup 2	
Subgroup 3			3 devices, c = 0
Destructive physical analysis	2101		
Subgroup 4			5 devices, c = 0
Thermal resistance	3161	See 4.5.2	
Subgroup 5 2/			
Barometric Pressure test	1001	Condition C, $V_{(ISO)} = V_{DS}$	5 devices, c = 0
2N7221		V_{DS} = 400 V dc	
2N7222		V_{DS} = 500 V dc	
Subgroup 6			5 devices, c = 0
Repetitive avalanche energy	3469	a. Peak current (I_{AR}): I_D. b. Peak gate voltage (V_{GS}): -10 V. c. Gate to source resistors: 25 ≤ R_{GS} ≤ 200. d. Temperature: T_J = 150°C +0°C, -10°C. e. Inductance: $(2 E_{AR}/(I_{D1})^2)$ x $((V_{BR} - V_{DD})/V_{BR})$ mH minimum. f. Number of pulses to be applied: 3.6×10^8. g. Supply voltage (V_{DD}): 50 V. h. Time in avalanche: 2 μs min., 20 μs max. i. Frequency: 500 Hz min.	
Electrical measurements		See table I, group A, subgroup 2	

1/ A separate sample for each test may be pulled.
2/ Not required for 2N7218, 2N7218U, 2N7219, and 2N7219U)
3/ Unless otherwise specified, inspections, conditions, and limits for "U" suffix devices are identical to non-"U" suffix devices.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.

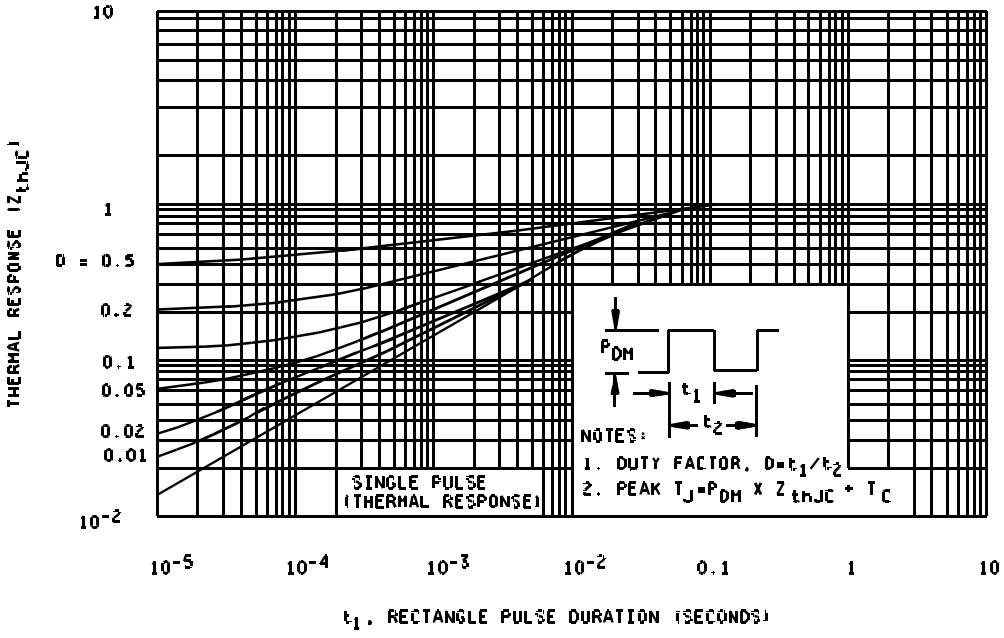


FIGURE 4. Thermal response.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.

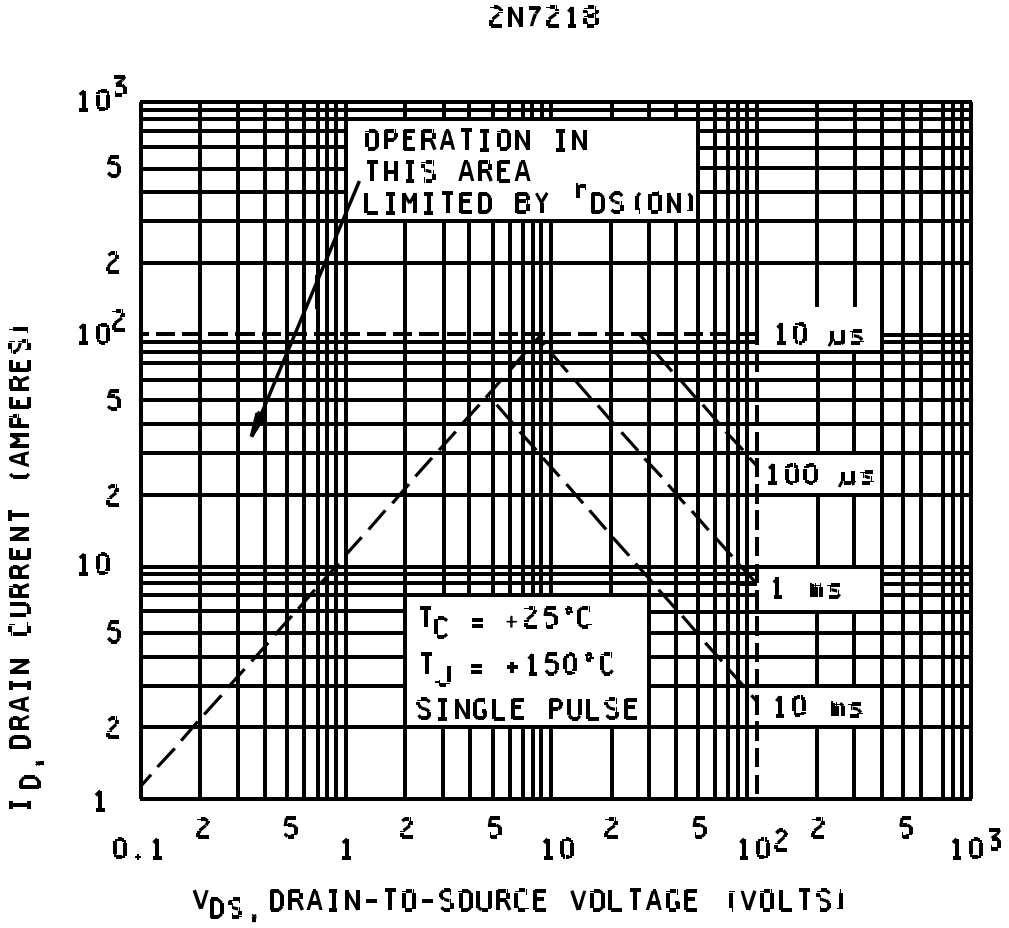


FIGURE 5. Safe operating area.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.

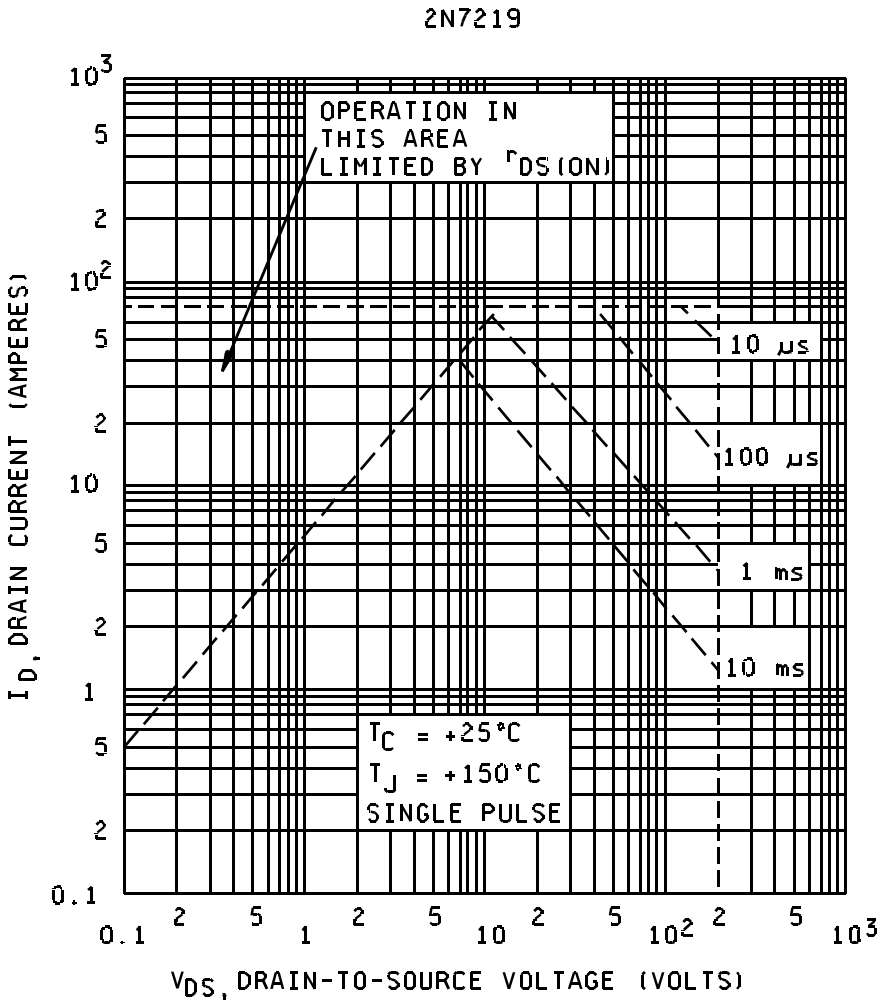


FIGURE 5. Safe operating area - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.

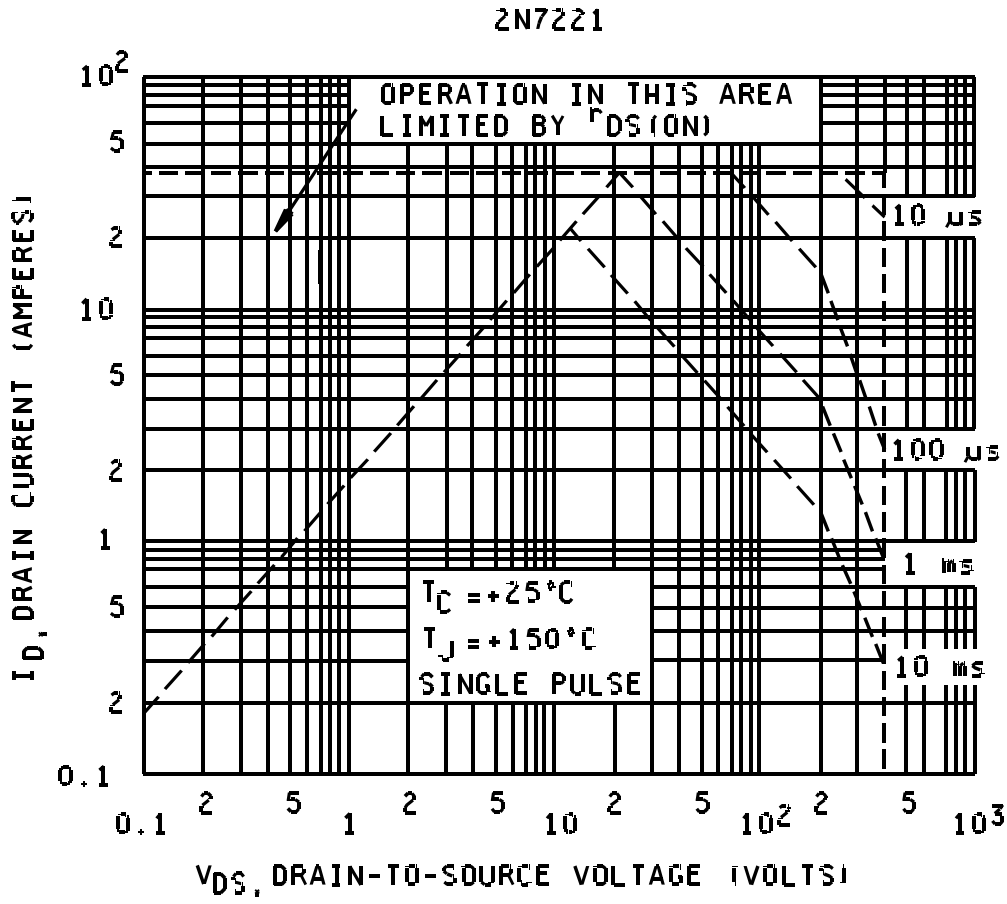


FIGURE 5. Safe operating area - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.

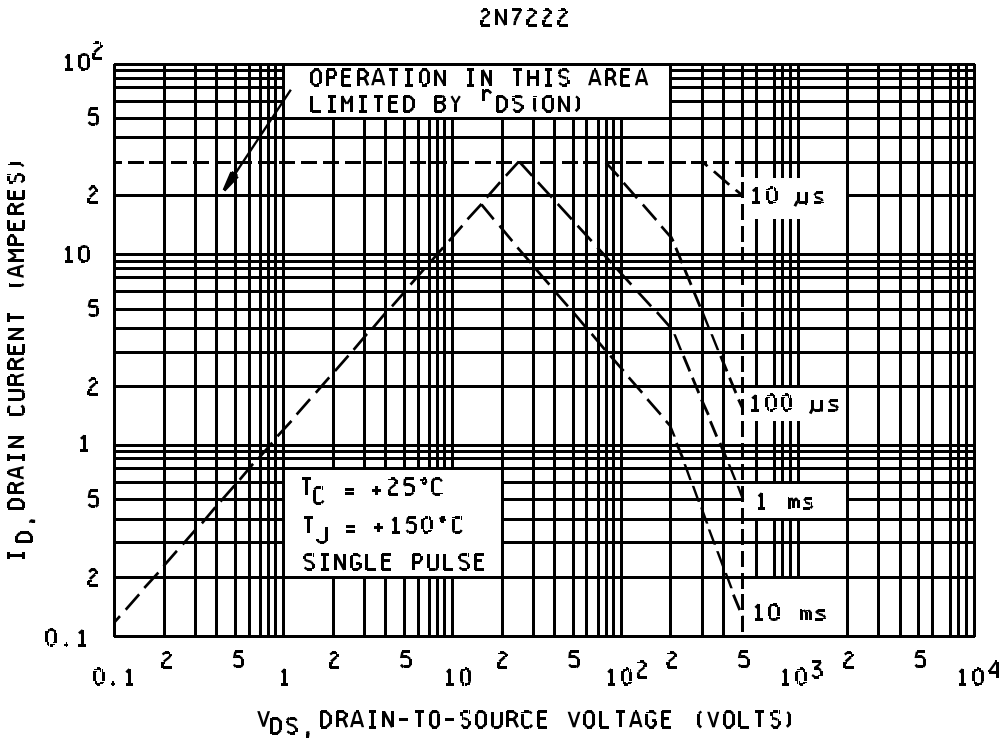


FIGURE 5. Safe operating area - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2.1).

b. Lead finish (see 3.3.1).

c. Type designation and product assurance level.

d. Packaging requirements (see 5.1).

e. For die acquisition, specify the JANHC or JANKC letter version (see figures 3, 4, and 5).

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, 3990 East Broad St., Columbus, OH 43216.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.

6.5 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's PIN. This information in no way implies that manufacturer's PIN's are suitable as a substitute for the military PIN.

Military PIN	Manufacturer's CAGE	Manufacturer's and user's PIN
2N7218	59993	IRFM 140
2N7219	59993	IRFM 240
2N7221	59993	IRFM 340
2N7222	59993	IRFM 440
2N7218U	59993	IRFN 140
2N7219U	59993	IRFN 240
2N7221U	59993	IRFN 340
2N7222U	59993	IRFN 440

6.6 Suppliers of JANHC and JANKC die. The qualified die suppliers with the applicable letter version (example JANHCA2N7218) will be identified on the QPL.

JANC ordering information		
	Manufacturer	
PIN	59993	
2N7218	JANHCA7218	
	JANKCA7218	
2N7219	JANHCA7219	
	JANKCA7219	
2N7221	JANHCA7221	
	JANKCA7221	
2N7222	JANHCA7222	
	JANKCA7222	

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 13, 19, 85, 99

Preparing activity:
DLA - ES

(Project 5961-1840)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/596D	**2. DOCUMENT DATE (YYMMDD)** 96/09/13

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, REPETITIVE AVALANCHE, FIELD EFFECT, TRANSISTOR, N-CHANNEL, SILICON, TYPES 2N7218, 2N7219, 2N7221, 2N7222, 2N7218U, 2N7219U, 2N7221U, 2N7222U, JANTX, JANTXV, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 614-692-0510 (2) AUTOVON 850-0510
c. ADDRESS (Include Zip Code) Defense Supply Center Columbus ATTN: DSCC-VAT 3990 East Broad St. Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:52Z
Check the source to verify that this is the current version before use.